Syllabus Details

Syllabus ID:	13769
Syllabus Name:	**Introduction to Software Engineering_Nhập môn kỹ thuật phần mềm**
Course Name English:	**Introduction to Software Engineering**
Subject Code:	**SWE202c**
Learning-Teaching Method:	Online, Active learning, Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) =82h online + 3h offline + 1h TE + 2h PE + 62 h self-study
Pre-Requisite:	PRO192
Description:	The course includes two parts: The first part introduces the use of Generative AI (GenAI) and Large Language Models (LLMs) in modern software development. Students will learn how AI-powered tools can accelerate the software engineering process, from analyzing requirements to generating code, writing tests, debugging, and producing technical documentation. The second part provides a comprehensive introduction to modern software engineering principles, methods, and practices. The course equips students with the foundational and advanced skills needed to design, develop, test, deploy, and maintain reliable software systems. Learners explore the full software development lifecycle (SDLC), including requirements engineering, architectural and object-oriented design, software construction, quality assurance, and DevOps practices. Emphasis is placed on practical, industry-relevant techniques such as version control, automated testing, API development, and iterative development using Agile. Through a combination of theoretical concepts, hands-on exercises, and real-world case studies, the course prepares students to work effectively in software engineering teams and build scalable, maintainable, and high-quality software products.
StudentTasks:	Student must get the certification of Software Engineering Specialization and Generative AI in Software Development course from Coursera to be accepted to the final examination
Tools:	
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1318/QĐ-ĐHFPT dated 11/27/2025
IsApproved:	**True**
Note:	1. Complete the online courses to be allowed to take Final Exam 2. Final Exam is included of Final Theory Exam (TE) & Final Practical Exam (PE): 100% 3. Students who complete all MOOCs before the course deadline will receive 1 bonus point; otherwise, no bonus point will be awarded. 4. Completion Criteria: Final TE Score >=4 & Final PE Score >=4 & FR >= 5 FR = min[10, (Final TE Score * weight of Final TE) + (Final PE Score * weight of Final PE) + bonus]
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	11/27/2025

4 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	MOOC 01: Generative AI in Software Development	Amazon	Coursera				True	False	True	https://www.coursera.org/learn/generative-ai-in-software-development#modules

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
2	MOOC 02: Software Engineering: Modeling Software Systems using UML	Kenneth W T Leung	Coursera				True	False	True	https://www.coursera.org/learn/software-engineering-modeling-software-systems-using-uml?specialization=software-engineering
3	MOOC 03: Software Engineering: Implementation and Testing	Kenneth W T Leung	Coursera				True	False	True	https://www.coursera.org/learn/software-engineering-implementation-and-testing?specialization=software-engineering
4	MOOC 04: Software Engineering: Software Design and Project Management	Kenneth W T Leung	Coursera				True	False	True	https://www.coursera.org/learn/software-engineering-software-design-and-project-management?specialization=software-engineering

4 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand and apply Generative AI models across the software development lifecycle—including code generation, testing, debugging, and documentation—while evaluating output quality, integrating GenAI into development workflows, and ensuring ethical, security, and intellectual-property compliance in real-world projects.
2	CLO2	understand software modeling principles and apply UML techniques to analyze requirements, design system structure and behavior, create and interpret UML diagrams, and evaluate models for accuracy, completeness, and consistency in real-world software projects.
3	CLO3	implement software components following industry coding standards, apply testing techniques to ensure software correctness and reliability, and evaluate code quality through systematic debugging, verification, and validation activities in real-world software projects.
4	CLO4	design software systems using appropriate design principles and patterns, plan and manage software projects effectively, coordinate team activities, monitor progress, and ensure project deliverables meet quality, schedule, and resource requirements in real-world settings

View mapping of CLOs to PLOs

Download All Student Material 12 sessions (45'/session)

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Offline 1: Introduction to the online course SWE202c. MOOC 1: Generative AI in Software Development Module 1: AI and building software	Offline,Online	CLO1	ITU	Course: Generative AI in Software Development https://www.coursera.org/learn/generative-ai-in-software-development		- See the syllabus of all 4 MOOCs - Participate in introduction slot (optional) - Watch the videos - Read the materials - Do all the quizzes. - Complete and submit all assignments.	
2	MOOC 1: Generative AI in Software Development Module 1: AI and building software (Cont.) Module 2: AI and Digital Transformation	Online	CLO1	U	Course: Generative AI in Software Development https://www.coursera.org/learn/generative-ai-in-software-development		- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
3	MOOC 1: Generative AI in Software Development Module 3 : Final project and course wrap up	Online	CLO1	U	Course: Generative AI in Software Development https://www.coursera.org/learn/generative-ai-in-software-development		- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
4	MOOC 2: Software Engineering: Modeling Software Systems using UML Module 1: Introduction to Software Engineering Module 2: Modeling Software Systems Using UML Module 3: Types of Relationships in Class Diagram	Online	CLO2	U	Course: Software Engineering: Modeling Software Systems using UML https://www.coursera.org/learn/software-engineering-modeling-software-systems-using-uml		- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
5	MOOC 2: Software Engineering: Modeling Software Systems using UML Module 4 : System Requirements Capture and Domain Modeling Module 5: Use Case Modeling	Online	CLO2	U	Course: Software Engineering: Modeling Software Systems using UML https://www.coursera.org/learn/software-engineering-modeling-software-systems-using-uml		- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
6	MOOC 2: Software Engineering: Modeling Software Systems using UML Module 6: Use Case Specification Module 7: Non-Functional Requirements	Online	CLO2	U	Course: Software Engineering: Modeling Software Systems using UML https://www.coursera.org/learn/software-engineering-modeling-software-systems-using-uml		- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
7	MOOC 3: Software Engineering: Implementation and Testing Module 1: Introduction to Software Development Module 2: Software Development Processes Module 3: Implementation and Defensive Programming	Online	CLO3	U	Course: Software Engineering: Implementation and Testing https://www.coursera.org/learn/software-engineering-implementation-and-testing		- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
8	MOOC 3: Software Engineering: Implementation and Testing Module 4: Debugging and Configuration Management Module 5: White Box Testing	Online	CLO3	U	Course: Software Engineering: Implementation and Testing https://www.coursera.org/learn/software-engineering-implementation-and-testing		- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
9	MOOC 3 :Software Engineering: Implementation and Testing Module 6: Black Box Testing Module 7: Acceptance Testing	Online	CLO3	U	Course: Software Engineering: Implementation and Testing https://www.coursera.org/learn/software-engineering-implementation-and-testing		- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
10	MOOC 4: Software Engineering: Software Design and Project Management Module 1: System Analysis and Design Module 2: State Machine Diagram	Online	CLO4	U	Course: Software Engineering: Software Design and Project Management https://www.coursera.org/learn/software-engineering-software-design-and-project-management		- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
11	MOOC 4: Software Engineering: Software Design and Project Management Module 3: Design Patterns Module 4: More Design Patterns	Online	CLO4	U	Course: Software Engineering: Software Design and Project Management https://www.coursera.org/learn/software-engineering-software-design-and-project-management		- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed)	
12	MOOC 4: Software Engineering: Software Design and Project Management Module 5: Software Quality Assurance Module 6: Managing Software Development	Offline,Online	CLO4	TU	Course: Software Engineering: Software Design and Project Management https://www.coursera.org/learn/software-engineering-software-design-and-project-management		- Watch all videos, read all materials in the MOOC - Finish quizzes, self-assessment, assignments and take part in discussion boards (if needed) Deadline of Coursera completion: End of Friday of the week. All students should participate in review slot (if any question)	

0 Constructive question(s)

2 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	PE	Final exam	1	50.0%	4	120'	CLO1, CLO2, CLO3, CLO4		5	All studied courses.	by Exam Board	Customized from the exercises of this specialization.
2	TE	Final exam	1	50.0%	4	60'	CLO1, CLO2, CLO3, CLO4	Multiple choices,Marked by Computer	50	All studied courses. Each module of course contributes 2-3 questions.	by Exam Board,using computer	Customized from the quizzes of this specialization.